

BISHOW LAMSAL

Machine Learning Engineer

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CAREER OBJECTIVE

Machine Learning Engineer with 3+ years of experience building and deploying AI systems across computer vision, NLP, and generative AI. Proven track record delivering end-to-end ML products — from OCR pipelines and agentic workflows to real-time traffic analytics — in production environments. Seeking to automate complex, high-value processes by leveraging LLMs, VLMs, and agentic AI architectures across fintech, healthcare, and autonomous systems.

TECHNICAL SKILLS

Languages & Databases	Python, SQL, MongoDB, Redis, Vector Database, MEM0
AI/ML Frameworks	TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, Hugging Face, Transformers
LLMs & Generative AI	LangChain, LangGraph, OpenAI, LLaMA, Mistral, Phi, Qwen, RAG
MLOps & Deployment	FastAPI, Flask, Streamlit, Docker, Jenkins, Git, Bitbucket, AWS, Nepali Cloud
Data & Visualization	Pandas, NumPy, Matplotlib, Seaborn, Plotly, TensorBoard, WandB
GPU & Edge	CUDA, TensorRT, cuDNN, Jetson

WORK EXPERIENCE

Machine Learning Engineer — Vertex Special Technology Jan 2024 – Present

- Led development of an agentic QA automation platform that reduces manual test case creation from hours to minutes, capable of generating hundreds of reusable action-based test scenarios without redundant AI calls — from a single user prompt using Playwright and LLM-driven UI agents.
- Built a GRPC-based chat interface that interprets user intent and dynamically selects between UI automation and test case generation workflows.
- Engineered an intelligent user story parser that extracts intent, classifies automation flows, maps dependencies, and concurrently executes multiple UI automation scenarios — generating structured test cases for each flow automatically.
- Designed a reusable DOM-caching strategy that eliminates redundant AI calls for repeat or modified test cases, significantly cutting automation cost and latency compared to conventional AI-heavy QA tools.
- Researched and prototyped an AI-powered visual comparison pipeline that captures live web pages via Playwright, extracts Figma design references, and uses computer vision to detect layout deviations, style inconsistencies, and component mismatches — eliminating manual design-to-implementation QA inspection
- Delivered an end-to-end agentic AI procurement system for the UN FAO that automates the full RFQ and ITB workflow — including document reading, data extraction, data entry, analysis, sheet generation, and Local Procurement Committee (LPC) report creation — enabling a single operator to handle a process that previously required multiple staff and extensive back-and-forth.
- Developed and deployed specialized OCR and Vision Language Models (VLMs) for the US financial sector, including custom data annotation templates for complex bordered and borderless table structures in bank statements.
- Collaborated with frontend, .NET backend, and cross-functional development teams to integrate AI outputs seamlessly via REST API and GRPC..

Machine Learning Engineer Associate — Bottle Technology Sep 2022 – Sep 2023

- Developed and deployed a real-time traffic analytics system across 4 cameras at 4 locations in Kathmandu and Lalitpur Metropolitan cities, capable of processing live footage at 20 FPS on Jetson Xavier edge devices.
- Implemented vehicle detection and tracking using YOLO and DeepSORT, including algorithms to detect one-way traffic violations and illegal highway stops using geometric and motion-based logic.
- Optimized model inference by converting to TensorRT format, significantly improving GPU utilization and achieving real-time 20 FPS throughput on edge hardware.
- Analyzed over 100 hours of recorded footage to deliver actionable traffic insights for Kathmandu and Lalitpur Metropolitan city authorities.
- Conducted real-time crowd management analysis at BNB Hospital and ticket counters, tracking and counting large volumes of individuals to support operational planning.
- Trained specialized OCR models for Devanagari script license plate recognition and ID card information extraction, enabling accurate reading of Nepali-language documents.
- Built and deployed end-to-end ML pipelines from model design to cloud integration using Docker, Jenkins, and AWS.

Machine Learning Trainee — Bottle Technology Jun 2022 – Aug 2022

- Studied and implemented classical image processing algorithms from scratch, both with and without OpenCV, based on research papers.
- Annotated training data, fine-tuned models, and tracked performance across experiments using TensorBoard and Weights & Biases (WandB).
- Researched and benchmarked multiple versions of YOLO for real-time object detection, comparing performance across speed and accuracy metrics.
- Explored supervised and unsupervised techniques for image color analysis, including in-depth study of RGB color spaces, grayscale conversion weight assignments, and image binarization.

ACADEMIC QUALIFICATION

Bachelor of Engineering in Computer Science — Visvesvaraya Technological University, Bangalore, India

CGPA: 7.53/10 | Aug 2016 – Dec 2020

PERSONAL PROJECTS

- **Healthcare Data Extraction** — Built a cloud-based pipeline to process ~1TB of split PDF health records, extracting key-value pairs and table data into structured JSON stored in MongoDB, with a Streamlit UI for patient data retrieval.
- **Personal Expense Tracker (Flutter/Kotlin/Python)** — Building a cross-platform mobile application that automatically extracts transaction data from bank SMS notifications using NLP-based parsing, categorizes income and expense flows, and visualizes spending patterns through interactive dashboards. Features include Chat interface, multi-bank account management, budget estimation, and AI-assisted development workflow using Claude Code.
- **Real-time Work Surveillance** — Developed employee activity monitoring system with real-time face detection and recognition using vector databases, tracking seated hours, mobile usage, and other behaviors with web-based visualization.
- **Face Detection, Recognition & Data Extraction** — Built a face similarity detection system combined with document classification and data extraction, deployed on Nepali Cloud to ensure data sovereignty.
- **Self-healing QA Automation** — Implemented Selenium-based QA automation with Generative AI to detect and adapt to UI template changes automatically.
- **Advanced E-Video-KYC** — Developed real-time liveness authentication system detecting facial movements (blink, mouth open, head turns) integrated with OCR-based document data extraction and voice assistant support.
- **News Recommendation System** — Built a hybrid recommendation engine using the MIND dataset combining content-based, collaborative filtering, and popularity-based approaches with Word2Vec embeddings and Multi-View Attention.

- **Captcha Recognition** — Built a captcha recognition system from scratch without relying on pre-trained models, implementing custom image preprocessing and classification pipelines.
- **Image Caption Generator** — Developed an automatic image captioning model combining CNN-based visual feature extraction with sequence generation using RNN/LSTM architecture..

CERTIFICATIONS & TRAINING

- Worked as mentor in Hack4SafeFoodHackathon
- 4-Month Data Science Training — Deerwalk Training Center
- 7-Week Web Development Internship — Rbits Technology
- Active Kaggle competitor — working with datasets from Kaggle and UCI Repository